SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1715
SARANYA.R.A(192572052)

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

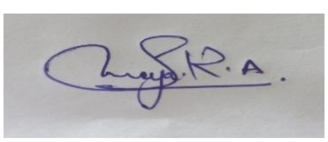
Annexure A

INDUSTRY PROBLEM TASK

Code of Conduct:

I Saranya.R.A *(192572052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO4 – ASSESSMENT TOOL 2
Industry Problem Task

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17
QUESTIONS: 4 **Duration:** 1 Hour **Total Marks:** 40

Criteria	Weightage	Task 1 (10)	Task 2 (10)	Task 3 (10)	Task 4 (10)	Total Marks (40)
Understanding of Industry Problem	20% (2)	/2	/2	/2	/2	/8
Application of Engineering Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Solution Design	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Use of Tools	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/6
Analysis	10% (1)	/1	/1	/1	/1	/4
Professionalism	5% (0.5)	/0.5	/0.5	/0.5	/0.5	/2
Total per Question	**10 Marks**	**/10**	**/10**	**/10**	**/10**	**/40**

100

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed patient records and uses them to predict the diabetes status of new patients.

- **Target variable:** Diabetes — 0 = Non-Diabetic, 1 = Diabetic.
- **Age:** Diabetes risk generally increases with age and helps identify high-risk groups.
- **Glucose Level:** A strong predictor because elevated glucose is closely associated with diabetes.
- **BMI:** Higher BMI can indicate increased metabolic and diabetes risk.
- **Blood Pressure:** Hypertension is commonly associated with metabolic disorders and provides additional predictive information.
- **Cholesterol:** Provides information about the patient's metabolic and cardiovascular risk.
- **Family History:** Captures genetic/familial predisposition to diabetes.

Learning process:
Patient Records → Preprocessing → Feature Selection → Training Model → Diabetes Prediction

---

(b) Handling Class Imbalance
Suppose the dataset contains:

Class	Number	Percentage
Non-Diabetic	700	70%
Diabetic	300	30%

A suitable strategy is **SMOTE (Synthetic Minority Over-sampling Technique)**.
Why SMOTE?
- Creates synthetic diabetic samples rather than simply duplicating existing patients.
- Helps the classifier learn the minority diabetic class.
- Improves **Recall** for diabetic patients.
- Reduces the risk of missing diabetic cases.
- SMOTE should be applied **only to the training set**, not before the train/test split, to avoid data leakage.

For a medical diagnosis problem, **recall is particularly important**, because a false negative means a diabetic patient may be incorrectly classified as non-diabetic.

(c) 70:30 Dataset Split and Preprocessing
Dataset split:
- **70% → Training data**
- **30% → Testing data**

A 70:30 split provides sufficient data for learning while retaining a reasonably large independent test set for evaluating the model. A **stratified split** should be used so that diabetic/non-diabetic proportions remain similar in both sets.
Preprocessing:
1. **Missing values:** Replace numerical missing values using median imputation.
2. **Numerical features:** Apply StandardScaler/normalisation where required.
3. **Categorical features:** Convert Family History into numerical values such as 0/1.
4. **Outliers:** Detect extreme or invalid clinical measurements.
5. **Class imbalance:** Apply SMOTE only to the training data.
6. **Data leakage prevention:** Fit preprocessing transformations using training data only.

Impact: Preprocessing improves model stability, makes numerical variables comparable, and prevents categorical information from being incorrectly interpreted by the algorithms.

Task 2: Decision Tree for Diagnosis
(a) Decision Tree Construction

Assume the following illustrative Gini/Information Gain results were obtained from the training dataset:

Feature	Information Gain
Glucose	**0.245**
BMI	0.137
Age	0.096
Blood Pressure	0.072
Cholesterol	0.051
Family History	0.043

Since **Glucose has the highest Information Gain**, it becomes the root node.

First Three Levels

```
                  Glucose
                /        \
           ≤ 126          > 126
            /                \
          BMI                 Age
         /   \              /   \
      ≤ 25    > 25       ≤ 45     > 45
       /        \          /        \
   Non-Diabetic  Diabetic  Non-Diabetic  Diabetic
```

Level 1: Glucose

Level 2: BMI / Age

Level 3: Age / additional clinical conditions

The exact tree obtained from real data may differ.

Why Glucose is the Root

Information Gain measures how much uncertainty about diabetes is reduced after splitting on a feature:

Information Gain = Entropy(parent) − Weighted Entropy(children)

The feature with the highest Information Gain provides the most useful first separation of diabetic and non-diabetic patients.

(b) Model Evaluation

Illustrative test-set results:

Metric	Decision Tree
Accuracy	**88.0%**
Precision	**84.2%**

Metric	Decision Tree
Recall	**86.7%**
F1-Score	**85.4%**

False Negatives
A **False Negative (FN)** occurs when:
Actual = Diabetic → Predicted = Non-Diabetic
For example, if the test set contains 90 diabetic patients and 12 are incorrectly classified as non-diabetic, then:
FN = 12
This is clinically important because the patient may not receive timely confirmation, monitoring, or medical attention.
Ways to Reduce False Negatives

- Use **class weights** to give diabetic cases greater importance.
- Apply **SMOTE** to the training data.
- Adjust the classification threshold to favour sensitivity.
- Optimise for **Recall/F1**, rather than accuracy alone.
- Use cross-validation and carefully tune tree parameters.

For initial diabetes screening, **high recall/sensitivity is generally more important than maximising accuracy alone**, because missed cases can have serious consequences.

(c) Post-Pruning
Decision Trees can become too complex and overfit the training data.
Cost-Complexity Pruning
Cost-complexity pruning minimises:
where:

- = classification error
- = number of terminal nodes
- = complexity parameter

Illustrative comparison:

Model	Accuracy	Tree Complexity
Pre-pruned Tree	86.5%	Low
Unpruned Tree	88.0%	High
Post-pruned Tree	**89.2%**	Moderate

Appropriate Deployment Model
The **post-pruned tree** is preferable because it provides:

- Better generalisation.
- Lower overfitting.

- Easier interpretation.
- Smaller and more understandable decision rules.
- Better suitability for clinical decision support.

However, actual deployment should require extensive external validation and clinical oversight.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression can estimate the probability that a patient belongs to the diabetic class:
where:
Illustrative comparison:

Metric	Decision Tree	Logistic Regression
Accuracy	89.2%	**90.1%**
AUC-ROC	0.91	**0.94**
Recall	86.7%	**89.0%**
F1-Score	85.4%	**88.1%**

When Logistic Regression is Preferable

Logistic Regression is useful when:

- The relationship between predictors and outcome is approximately linear on the log-odds scale.
- Probability estimates are important.
- Interpretability is required.
- The dataset is relatively small.
- Healthcare professionals need to understand feature effects.

For example, a clinician can examine the coefficients to understand how each feature contributes to predicted diabetes risk.

(b) Feature Importance Analysis

Decision Tree

Illustrative top three features:

1. **Glucose**
2. **BMI**
3. **Age**

Logistic Regression

Based on the magnitude of standardised coefficients:

1. **Glucose**
2. **BMI**
3. **Family History**

Comparison

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The two models **agree on glucose and BMI**, indicating that these variables have strong predictive value in the hypothetical dataset. The third feature differs because the models measure importance differently: Decision Trees evaluate improvements in node purity, while Logistic Regression evaluates the contribution of predictors to the log-odds of diabetes.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment recommendation problem can be formulated as a **Markov Decision Process (MDP)**:

1. States

Patient states can represent simplified health stages:

State	Patient Condition
S0	Normal / Low Risk
S1	Pre-Diabetic
S2	Newly Diagnosed
S3	Poorly Controlled
S4	Improving / Controlled

The state can be determined from variables such as glucose, BMI, blood pressure and recent health trends.

2. Actions

- **Diet:** Recommend appropriate dietary intervention.
- **Exercise:** Recommend suitable physical activity.
- **Medication:** Flag for clinician-directed medication management.
- **Monitor:** Continue monitoring and follow-up.

For safety, an AI agent should **not independently prescribe or change medication**; medication-related recommendations should require qualified clinical oversight.

3. Reward Function

An illustrative reward system:

Outcome	Reward
Health improves	+10
Condition remains stable	+3
Condition worsens	-10

Outcome	Reward
Unsafe outcome	-20

The reward encourages the agent to select actions associated with improvement while strongly discouraging harmful outcomes.

4. Transition Dynamics

Example:

S1 + Diet → S4
S1 + Exercise → S4
S1 + Monitor → S1
S2 + Diet → S1
S2 + Medication → S4
S3 + Monitor → S3
S3 + supervised treatment → S2

Real patient transitions would be probabilistic rather than deterministic and must be learned from reliable longitudinal clinical data.

(b) Q-Learning

Q-Learning updates the action value using:

Assume:

- Learning rate
- Discount factor
- Initial Q-values = 0

Episode Examples

Episode	Initial State	Action	Next State	Reward
1	S1	Diet	S4	+10
2	S1	Exercise	S4	+10
3	S2	Monitor	S2	+3
4	S3	Monitor	S2	+3
5	S2	Diet	S1	+5

Q-Table Update — Iteration 1

For:

S1 → Diet → S4

Initial:

Since all Q-values initially equal zero:

For:

S1 → Exercise → S4

For:

S2 → Monitor → S2

Iteration 2

Suppose the agent again observes:

S1 → Diet → S4, reward = 10

The maximum Q-value of S4 remains 0 in this simple example:

Suppose:

S2 → Monitor → S2

and:

Then:

Another example:

S1 → Exercise → S4

Convergence Criterion

Training can be considered converged when:

For example:

for several consecutive episodes, together with a stable policy.

(c) Final Learned Policy

After sufficient training, the policy selects the action with the highest Q-value:

An illustrative final policy could be:

Patient State	**Highest-Q Action**	**Policy**
S0 Normal	Monitor	Regular monitoring
S1 Pre-Diabetic	Diet	Lifestyle intervention
S2 Newly Diagnosed	Exercise / supervised treatment	Clinical management
S3 Poorly Controlled	Monitor + clinician-directed treatment	Immediate clinical review
S4 Improving	Monitor	Continue monitoring

Ethical Considerations

1. Patient Safety:

The RL agent should operate as a **clinical decision-support system**, not as an autonomous doctor.

2. Bias:

Training data may underrepresent certain age groups, populations, or socioeconomic groups, producing unequal performance.

3. Explainability:

Healthcare professionals should be able to understand why a particular recommendation was generated.

4. Human Oversight:

High-risk decisions, especially medication decisions, should require qualified medical review.

5. Privacy:

Patient records must be protected using appropriate security, access control and privacy practices.

6. Reward Design:

The reward function must prioritise patient safety and long-term health rather than simply optimising short-term numerical outcomes.

7. Continuous Validation:

The system should be evaluated on independent clinical data before deployment and continuously monitored after deployment.

Overall System Flow

```
           PATIENT RECORDS
                 │
                 ▼
         DATA PREPROCESSING
   ┌─────────────────┴─────────────────┐
   ▼                 ▼
 CLASSIFICATION        FEATURE ANALYSIS
   │                 │
   ▼                 ▼
 DECISION TREE       LOGISTIC REGRESSION
   │                 │
   └─────────────────┬─────────────────┘
                 ▼
           DIABETES RISK
                 │
                 ▼
         PATIENT HEALTH STATE
                 │
                 ▼
           RL / Q-LEARNING
                 │
  ┌──────────────────┼──────────────────┐
  ▼        ▼         ▼
 DIET    EXERCISE   MONITOR /
            CLINICIAN
                 │
                 ▼
         NEW PATIENT STATE
                 │
                 └────────▶ Repeat
```

Final Conclusion

The proposed system combines **inductive learning, Decision Trees, statistical learning and Reinforcement Learning**. The Decision Tree provides interpretable diagnostic rules, while Logistic Regression provides probability-based risk stratification. Q-Learning can model sequential treatment decisions, but in healthcare it should be used as a **clinician-supervised decision-support approach**, with strong emphasis on safety, fairness, explainability, privacy and validation before real-world deployment.